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COM232

HEIGHT AND WEIGHT DATASET

PERSON	HEIGHT (CM)	WEIGHT (Kg)	r_{women}	r_{men}
A	158	52	0.98	0.02
B	162	56	0.95	0.05
C	166	60	?	?
D	175	72	0.10	0.90
E	180	78	0.05	0.95
F	185	84	0.02	0.98

PARAMETER	WOMEN	MEN
mean (μ)	$\begin{bmatrix}160\\55\end{bmatrix}$	$\begin{bmatrix}180\\78\end{bmatrix}$
MIXING COEFFICIENT (π)	0.5	0.5
COVARIANCE (Σ)	$\begin{bmatrix}9 & 0\\0 & 16\end{bmatrix}$	$\begin{bmatrix}16 & 0\\0 & 25\end{bmatrix}$

i) COMPUTE THE RESPONSIBILITY FOR SAMPLE C.

$$N(C \mid \mu_{women}, \Sigma_{women}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}\left(\frac{(x_h-\mu_h)^2}{9} + \frac{(x_w-\mu_w)^2}{16}\right)\right)$$

$|\Sigma|^{1/2} = \sqrt{9\times16}$
$= 12$

$$= \frac{1}{75.398} \exp\left(-\frac{1}{2}\left(\frac{(166-160)^2}{9} + \frac{(60-55)^2}{16}\right)\right)$$

$$r_{women} = \frac{1}{75.398} \exp(-2.782) = \boxed{8.212\times10^{-4}}$$

$$N(C \mid \mu_{men}, \Sigma_{men}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}\left(\frac{(x_h-\mu_h)^2}{16} + \frac{(x_w-\mu_w)^2}{25}\right)\right)$$

$|\Sigma|^{1/2} = \sqrt{16\times25}$
$= 20$

$$= \frac{1}{125.664} \exp\left(-\frac{1}{2}\left(\frac{(166-180)^2}{16} + \frac{(60-78)^2}{25}\right)\right)$$

$$= \frac{1}{125.664} \exp\left(-\frac{1}{2}(25.21)\right) = \frac{1}{125.664} \exp(-12.605)$$

$$r_{men} = \boxed{2.64\times10^{-8}}$$

COMPUTE RESPONSIBILITIES

$$r_{ic} = \frac{\pi_c \, N(x_i \mid \mu_c, \Sigma_c)}{\sum_{j=1}^{K} \pi_j \, N(x_i \mid \mu_j, \Sigma_j)}$$

$$r_{women} = \frac{0.5 \times 8.212\times10^{-4}}{0.5 \times 8.212\times10^{-4} + 0.5 \times 2.67\times10^{-8}} = 0.9996 = \boxed{1}$$

$$r_{men} = \frac{0.5 \times 2.67\times10^{-8}}{0.5 \times 8.212\times10^{-4} + 0.5 \times 2.67\times10^{-8}} = 0.00003 = \boxed{0}$$

2) COMPUTE THE DATA POINTS FOR EACH CLUSTER

$WOMEN = 1+1+1+0+0+0 = \boxed{3}$ $\quad$ $MEN \; 0+0+0+1+1+1 = \boxed{3}$

3) COMPUTE THE UPDATED MIXING COEFFICIENT FOR EACH CLUSTER.

$\pi_{WOMEN} = \frac{3}{6} = \frac{1}{2} = \boxed{0.5}$ $\quad$ $\pi_{MEN} = \frac{3}{6} = \frac{1}{2} = \boxed{0.5}$

4) COMPUTE THE UPDATED MEANS OF EACH CLUSTER.

$\mu_{height} = \frac{158+162+166}{3} = \boxed{162}$ $\quad$ $\mu_{height} = \frac{175+180+185}{3} = \boxed{180}$

$\mu_{weight} = \frac{52+56+60}{3} = \boxed{56}$ $\quad$ $\mu_{weight} = \frac{72+78+84}{3} = \boxed{78}$

$\boxed{\mu_{women} = \begin{bmatrix}162\\56\end{bmatrix}}$ $\quad$ $\boxed{\mu_{men} = \begin{bmatrix}180\\78\end{bmatrix}}$

5) COMPUTE THE UPDATED COVARIANCE OF EACH CLUSTER.

$\mu_{women} = \begin{bmatrix}162\\56\end{bmatrix} = A=\begin{bmatrix}158\\52\end{bmatrix} \; B=\begin{bmatrix}162\\56\end{bmatrix} \; C=\begin{bmatrix}166\\60\end{bmatrix}$

$\mu_{men} = \begin{bmatrix}180\\78\end{bmatrix} = D\begin{bmatrix}175\\72\end{bmatrix} \; E=\begin{bmatrix}180\\78\end{bmatrix} \; F=\begin{bmatrix}185\\84\end{bmatrix}$

$(A-\mu_{women})\left(\begin{bmatrix}162\\56\end{bmatrix}-\begin{bmatrix}158\\52\end{bmatrix}\right)=\begin{bmatrix}4\\4\end{bmatrix} \; (A-\mu_w)^T=[4,4]$

$A_{W} = \begin{bmatrix}4\\4\end{bmatrix}-[4,4] = \begin{bmatrix}16 & 16\\16 & 16\end{bmatrix}$

$B_{W} = \left(\begin{bmatrix}162\\56\end{bmatrix}-\begin{bmatrix}162\\56\end{bmatrix}\right)=\begin{bmatrix}0\\0\end{bmatrix}-[0,0]=[0]$

$C_{W} = \left(\begin{bmatrix}162\\56\end{bmatrix}-\begin{bmatrix}166\\60\end{bmatrix}\right)=\begin{bmatrix}-4\\-4\end{bmatrix}-[-4,-4]=\begin{bmatrix}16 & 16\\16 & 16\end{bmatrix}$

$\Sigma_W = \frac{1}{3}\begin{bmatrix}16 & 16\\16 & 16\end{bmatrix}+0+\begin{bmatrix}16 & 16\\16 & 16\end{bmatrix}$

$= \frac{1}{3}\begin{bmatrix}32 & 32\\32 & 32\end{bmatrix} = \boxed{\begin{bmatrix}10.67 & 10.67\\10.67 & 10.67\end{bmatrix}}$

$D_M = \left(\begin{bmatrix}175\\72\end{bmatrix}-\begin{bmatrix}180\\78\end{bmatrix}\right)=\begin{bmatrix}-5\\-6\end{bmatrix}-[-5,-6]=\begin{bmatrix}25 & 30\\30 & 36\end{bmatrix}$

$E_M = \left(\begin{bmatrix}180\\78\end{bmatrix}-\begin{bmatrix}180\\78\end{bmatrix}\right)=\begin{bmatrix}0\\0\end{bmatrix}-[0,0]=[0]$

$F_M = \left(\begin{bmatrix}185\\84\end{bmatrix}-\begin{bmatrix}180\\78\end{bmatrix}\right)=\begin{bmatrix}5\\6\end{bmatrix}-[5,6]=\begin{bmatrix}25 & 30\\30 & 36\end{bmatrix}$

$\Sigma_M = \frac{1}{3}\begin{bmatrix}25 & 30\\30 & 36\end{bmatrix}+0+\begin{bmatrix}25 & 30\\30 & 36\end{bmatrix} = \frac{1}{3}\begin{bmatrix}50 & 60\\60 & 72\end{bmatrix} = \boxed{\begin{bmatrix}16.67 & 20\\20 & 24\end{bmatrix}}$